



MS/MA IN ECONOMICS

DYNAMIC STOCHASTIC MODELS

Lecture 2

OLG Model

Tomás E. Baigorria

Martín A. Fradkin

Nicolás P. Monteverde Capello

Professor: George McCandless

July 1, 2026

1. Budget constraints in a three-period OLG economy

We extend the class environment so that each generation lives for *three* periods instead of two. A member of the cohort born in period t is alive in periods t , $t + 1$ and $t + 2$. The agent supplies labor in the first two periods of life (when “young” and “middle-aged”) and is retired in the third (when “old”):

$$h_t(t) \geq 0, \quad h_t(t + 1) \geq 0, \quad h_t(t + 2) = 0.$$

As in the class model, labor endowments are supplied inelastically, capital saved at the end of a period is used in production the following period and depreciates fully within that period, and agents may also lend to / borrow from members of their own cohort. We keep the class notation:

- $c_t(s)$: consumption in period s of an agent born in t ;
- w_s : wage per unit of labor in period s ;
- $k_t(s)$: capital carried by the cohort into period s (bought at the end of $s - 1$);
- $l_t(s)$: net lending of the cohort in period s (negative if borrowing);
- r_s : gross return on private loans made in s (repaid in $s + 1$);
- $rental_s$: rental rate of capital used in period s .

We drop the household superscript h because agents within a cohort are identical; individual variables coincide with per-capita variables.

Budget constraints of the cohort born at t

In each period the cohort splits its available resources between consumption, new lending and the purchase of capital for the next period.

Period t — young, works. The only resource is labor income; it is allocated to consumption, lending and capital for $t + 1$:

$$w_t h_t(t) = c_t(t) + l_t(t) + k_t(t + 1). \quad (1a)$$

Period $t + 1$ — middle-aged, works. The cohort now has labor income *plus* the returns on the loans and capital chosen in t . These are spent on consumption, new lending and capital for $t + 2$:

$$w_{t+1} h_t(t + 1) + r_t l_t(t) + rental_{t+1} k_t(t + 1) = c_t(t + 1) + l_t(t + 1) + k_t(t + 2). \quad (1b)$$

Period $t + 2$ — old, retired. There is no labor income. The agent consumes the returns on the assets accumulated in $t + 1$ and leaves nothing behind (no bequests):

$$r_{t+1} l_t(t + 1) + rental_{t+2} k_t(t + 2) = c_t(t + 2). \quad (1c)$$

Same cohort written for $t = 1, 2, 3$

Taking the cohort born in period 1 (so its three periods of life are the calendar periods 1, 2, 3) the three budget constraints read

$$t = 1 : \quad w_1 h(1) = c(1) + l(1) + k(2), \quad (2a)$$

$$t = 2 : \quad w_2 h(2) + r_1 l(1) + rental_2 k(2) = c(2) + l(2) + k(3), \quad (2b)$$

$$t = 3 : \quad r_2 l(2) + rental_3 k(3) = c(3), \quad (2c)$$

where $h(1), h(2)$ are the (given) labor endowments supplied while working and $h(3) = 0$.

No-arbitrage and the lifetime budget constraint

With perfect foresight, loans and capital are perfect substitutes as stores of value, so within each holding period their gross returns must coincide:

$$r_1 = \text{rental}_2, \quad r_2 = \text{rental}_3. \quad (3)$$

Using (3) to substitute out $l(1)$ and $l(2)$ collapses (2a)–(2c) into a single lifetime budget constraint: the present value of lifetime consumption equals the present value of lifetime labor income,

$$\boxed{c(1) + \frac{c(2)}{r_1} + \frac{c(3)}{r_1 r_2} = w_1 h(1) + \frac{w_2 h(2)}{r_1}}. \quad (4)$$

Income earned in the third period is zero because the cohort is retired, so the right-hand side only contains the wage income of the first two periods, discounted back to period 1 at the sequence of equilibrium returns.

2. Exercise 1: a lump-sum transfer from young to old

Find the equilibrium in a constant-population economy where, in each period, the government levies a small lump-sum tax τ on each young agent and hands that amount to each old agent. Both groups treat τ as lump sum. Under what conditions does the tax raise welfare, and under what conditions does it not?

We return to the two-period class model (full depreciation). Population is constant, $N(t) = N$, so in every period there are N young and N old. The government collects $N\tau$ from the young and distributes it to the N old, so each old agent receives exactly τ .

Equilibrium

The transfer enters the budget constraints of the cohort born in t as:

$$\text{young: } w_t h_t(t) - \tau = c_t(t) + l_t(t) + k_t(t+1), \quad (5a)$$

$$\text{old: } c_t(t+1) = w_{t+1} h_t(t+1) + r_t l_t(t) + \text{rental}_{t+1} k_t(t+1) + \tau. \quad (5b)$$

With the no-arbitrage condition $r_t = \text{rental}_{t+1}$, combining (5a)–(5b) gives the lifetime budget constraint

$$c_t(t) + \frac{c_t(t+1)}{r_t} = \underbrace{w_t h_t(t) + \frac{w_{t+1} h_t(t+1)}{r_t}}_{\text{wage wealth } W_t} - \tau \left(1 - \frac{1}{r_t}\right). \quad (6)$$

The whole effect of the scheme on a given cohort is the term $-\tau \left(1 - \frac{1}{r_t}\right) = -\tau \frac{r_t - 1}{r_t}$: the agent pays τ when young and is repaid τ when old, and the *implicit gross return* of this forced “saving” is exactly

$$\frac{\tau \text{ received}}{\tau \text{ paid}} = 1 = 1 + n,$$

the gross population growth rate (here $n = 0$). This is the defining feature of a pay-as-you-go scheme: its return is the growth rate of the tax base, not the market interest rate.

To close the model, optimal saving of the young (Cobb–Douglas / log utility, labor only when young for transparency) is

$$s_t = \frac{\beta}{1 + \beta} (w_t - \tau) - \frac{1}{1 + \beta} \frac{\tau}{r_t},$$

so that the aggregate law of motion of capital becomes

$$K(t+1) = Ns_t = \underbrace{\frac{\beta}{1+\beta} N(w_t - \tau)}_{\text{lower disposable income}} - \underbrace{\frac{1}{1+\beta} \frac{N\tau}{r_t}}_{\text{future transfer crowds out saving}}. \quad (7)$$

Both terms are decreasing in τ : the tax lowers disposable income when young and the promised transfer reduces the motive to save privately. The transfer therefore crowds out capital: $K^*(\tau) < K^*(0)$, and the equilibrium is the steady state of (7) together with the factor prices $w_t = F_H(K(t), H)$, $rental_t = F_K(K(t), H)$ and $r_t = rental_{t+1}$.

Welfare: when does the transfer help?

The household effectively chooses between two stores of value with different returns:

private capital: gross return $r = rental = F_K$, transfer scheme: implicit gross return $1 + n = 1$.

Whether forcing some saving through the scheme is desirable depends on which return is larger — equivalently, on the sign of $r - 1$ in (6):

The lump-sum transfer raises welfare if and only if the equilibrium gross interest rate is below the (gross) population growth rate, i.e.

$$r = F_K < 1 + n = 1.$$

It *reduces* welfare if $r > 1$, and is neutral (to first order) at the golden rule $r = 1$. With full depreciation and constant population, the golden-rule capital stock maximizes steady-state consumption $C = F(K, H) - K$ and satisfies $F_K = 1$, i.e. $r = 1$.

- **Dynamic inefficiency ($r < 1$, capital above the golden rule).** The economy is over-accumulating: the marginal unit of capital returns less than the $1+n$ offered by the transfer scheme. Shifting resources from private capital to the pay-as-you-go scheme raises the return earned on the young's "saving" and *can make every generation better off simultaneously* — a Pareto improvement. In (6) the wealth effect $-\tau \frac{r-1}{r}$ is positive because $r < 1$. *The tax improves welfare.*
- **Dynamic efficiency ($r > 1$, capital below the golden rule).** Now private capital earns more than the scheme's implicit return. The transfer crowds out high-return capital and leaves the representative agent with lower lifetime wealth ($-\tau \frac{r-1}{r} < 0$). No Pareto improvement is possible. *The tax reduces welfare.*
- **Golden rule ($r = 1$).** The two returns coincide; the scheme is welfare-neutral to first order.

The condition is the standard Samuelson–Diamond result: an unfunded transfer from young to old is beneficial exactly when the economy is dynamically inefficient. The reason a small *lump-sum* scheme can yield a first-order welfare gain is precisely that it does not distort the consumption–saving margin directly; it merely substitutes a return of $1 + n$ for a market return r , which is an improvement only when $r < 1 + n$.

3. Exercise 2: partial depreciation $\delta < 1$

Solve the class model but with depreciation $\delta < 1$ (e.g. $\delta = 0.5$). Is stationary-state output higher or lower? Is stationary-state utility higher or lower? Starting from K_0 below the stationary state, is growth faster or slower?

What changes when capital does not fully depreciate

If only a fraction δ of capital is used up in production, the old keep the undepreciated remainder $(1 - \delta)k_t(t + 1)$ and sell it to the young. In a one-good economy a unit of undepreciated capital is worth one unit of the consumption good, so the gross payoff of a unit of capital saved by the young is the rental *plus* its resale value. The budget constraints of cohort t become

$$\text{young: } w_t h_t(t) = c_t(t) + l_t(t) + k_t(t + 1), \quad (8a)$$

$$\text{old: } c_t(t + 1) = w_{t+1} h_t(t + 1) + r_t l_t(t) + [\text{rental}_{t+1} + (1 - \delta)] k_t(t + 1). \quad (8b)$$

No-arbitrage between loans and capital now equates the loan return to the *total* return on capital,

$$\boxed{r_t = \text{rental}_{t+1} + (1 - \delta)} = F_K(K(t+1), H) + (1 - \delta), \quad (9)$$

and the feasibility constraint carries the surviving capital forward,

$$F(K(t), H) + (1 - \delta)K(t) = C(t) + K(t + 1). \quad (10)$$

Relative to the class benchmark ($\delta = 1$, where $r_t = \text{rental}_{t+1}$ and all capital must be replaced every period), the only structural change is the extra $(1 - \delta)$ in the return (9) and in the resource constraint (10).

Law of motion and stationary state

With Cobb–Douglas utility $u = c_t(t) c_t(t + 1)^\beta$ and $Y = K^\theta H^{1-\theta}$, saving of the young is

$$s_t = \frac{\beta}{1 + \beta} w_t h_t(t) - \frac{1}{1 + \beta} \frac{w_{t+1} h_t(t + 1)}{R_t}, \quad R_t \equiv \text{rental}_{t+1} + (1 - \delta), \quad (11)$$

and aggregate saving equals next period's capital, $K(t + 1) = N s_t$, giving a law of motion $K(t + 1) = G(K(t); \delta)$ whose stationary state K^* solves $K^* = G(K^*; \delta)$.

The key comparative static is transparent from (11). A lower δ raises the gross return R_t (more of the capital survives to be resold). This shrinks the present value of old-age labor income $w_{t+1} h_t(t + 1)/R_t$, so for any given $K(t)$ desired saving — and hence $K(t + 1)$ — is *higher*. The G schedule shifts up, and its intersection with the 45° line moves to a *higher* K^* .

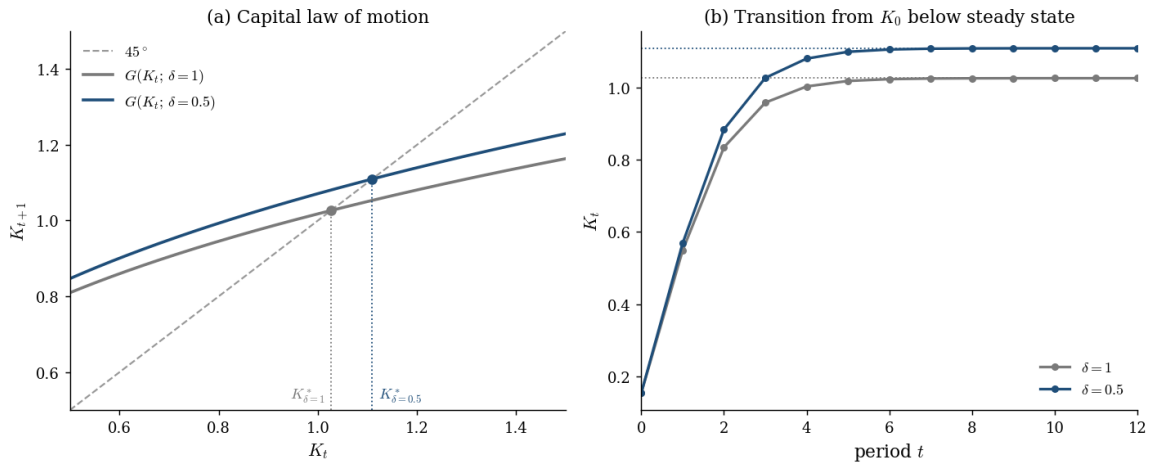


Figure 1: Effect of lower depreciation in the class economy (Cobb–Douglas utility and technology, second-period labor income, $\theta = 0.33$).

The capital law of motion $K_{t+1} = G(K_t; \delta)$ shifts up as δ falls, because the higher gross return $R = F_K + (1 - \delta)$ raises saving; its intersection with the 45° line moves to a higher K^* . From a

common initial K_0 below both stationary states, accumulation under $\delta = 0,5$ is faster and settles at a higher level.

Stationary output is therefore *higher* than under full depreciation: since $Y^* = (K^*)^\theta H^{1-\theta}$ is increasing in K^* , the higher fixed point carries output up with it. The economic content is that a more durable capital stock lets the same flow of saving sustain a larger stock. Stationary utility is also *higher* (as long as the economy remains dynamically efficient), because two forces push in the same direction. On one hand the wage $w^* = (1-\theta)(K^*/H)^\theta$ increases with K^* , enlarging the first-period budget; on the other, each unit saved now pays the higher gross return $R^* = F_K + (1-\delta)$, enlarging old-age consumption. The lifetime budget set expands on both margins, so welfare improves. Finally, starting from some K_0 below the stationary state, growth is *faster*: because a smaller fraction of the stock is destroyed each period, net accumulation is quicker, the G schedule lies above the full-depreciation one at every $K(t)$, the growth factor $K(t+1)/K(t)$ is larger throughout the transition, and the economy climbs toward a higher target. Figure 1 illustrates both effects: panel (a) shows the law of motion shifting up and the steady state moving right as δ falls from 1 to 0,5, and panel (b) shows the corresponding transition from a common K_0 , which is both steeper and converges to a higher level.

These conclusions hinge on a feature of the class specification that is worth confirming against the slides. Under logarithmic (Cobb–Douglas) preferences the saving rate out of *first-period* wage income is the constant $\frac{\beta}{1+\beta}$, independent of the return. If households worked only when young ($h_t(t+1) = 0$), the return-bearing term in (11) would vanish, the law of motion would reduce to $K(t+1) = \frac{\beta}{1+\beta} Nw_t$ regardless of δ , and stationary K^* , output and the transition path would all be *unchanged*; the only effect of a lower δ would then be a higher return R^* and hence strictly higher utility, with output and growth unaffected. The “higher output / faster growth” answers rest on the class example’s second-period labor income $h_t(t+1) > 0$, which is precisely what makes saving, and therefore accumulation, respond to the return. It should also be noted that the utility comparison is favorable only while the economy stays dynamically efficient: if the larger K^* pushed it past the golden rule ($F_K < 1$), additional capital would reduce steady-state consumption — the over-accumulation case in which the transfer of Exercise 1 would itself be welfare improving.